

SIMATS ENGINEERING

ASSESSMENT TOOL 1

ANALYTICAL PROBLEM SOLVING

COURSE NAME: Artificial Intelligence

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Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one.

Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug?

Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Given:

- **4-gallon jug** → capacity = 4 gallons
- **3-gallon jug** → capacity = 3 gallons
- No measurement markings.

Goal:

- Get **exactly 2 gallons in the 4-gallon jug**.

Procedure:

1. Fill the **3-gallon jug** completely.
2. Pour the water from the **3-gallon jug into the 4-gallon jug**.
 - 4-gallon = 3
 - 3-gallon = 0
3. Fill the **3-gallon jug** again.
4. Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full.
 - 4-gallon = 4
 - 3-gallon = 2
5. Empty the **4-gallon jug** completely.

- 4-gallon = 0
 - 3-gallon = 2
6. Pour the remaining **2 gallons** from the 3-gallon jug into the 4-gallon jug.
 7. Now the **4-gallon jug contains exactly 2 gallons**.

Final State:

4-gallonjug = 2

3-gallon jug = **0 gallons**

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

- i. What are the percept's for this agent?
- ii. Characterize the operating environment.
- iii. What are the actions the agent can take?
- iv. How can one evaluate the performance of the agent?
- v. What sort of agent architecture do you think is most suitable for this agent?

Why?

percepts for this agent

- Images and videos of the Martian surface.
- Temperature and atmospheric pressure.
- Distance from rocks, obstacles, and other objects.
- Soil and rock composition.
- Chemical properties of collected samples.
- Rover's position and location.
- Battery level and power availability.
- Wheel movement and rover speed.
- Signs of water, ice, or other interesting materials.

ii. Characterize the operating environment

The Mars Rover operates in an environment that is:

- **Partially observable** – The rover cannot observe the entire Mars surface.
- **Unknown** – It may encounter previously unexplored terrain.
- **Dynamic** – Weather, dust, lighting, and other conditions can change.
- **Sequential** – One action can affect future actions.
- **Continuous** – Movement, distance, temperature, and other values vary continuously.
- **Deterministic/uncertain** – Some actions may not produce exactly the expected result.
- **Single-agent** – The rover mainly operates independently, although it communicates with Earth.
- **Inaccessible and dangerous** – Human intervention is limited because of the large distance from Earth.

iii. What are the actions the agent can take

- Move forward and backward.
- Turn left or right.
- Stop.
- Avoid obstacles.
- Take photographs.
- Scan rocks and soil.
- Collect samples.
- Analyse samples using scientific instruments.
- Drill into the Martian surface.
- Store samples.
- Send data and images to Earth.
- Navigate to a selected location.

iv. How can one evaluate the performance of the agent

- Amount of useful scientific data collected.
- Number of samples successfully collected and analysed.
- Accuracy of sample analysis.

- Distance successfully explored.
- Number of important geological discoveries.
- Ability to avoid obstacles and hazards.
- Efficient use of battery/power.
- Successful communication with Earth.
- Accuracy in navigation.

Performance measure:

A good Mars Rover should **explore maximum useful area, collect high-quality samples and scientific data, avoid damage, use resources efficiently, and successfully transmit the results to Earth.**

What sort of agent architecture is most suitable

- The rover cannot observe the complete environment, so it needs an internal model of Mars.
- It has specific goals such as exploring an area, collecting samples, and analysing rocks.
- It must choose the best action among several possible actions.
- It needs to consider battery power, safety, distance, scientific importance, and time.
- It can update its internal model when it receives new sensor information.
- It can make decisions independently when communication with Earth is delayed.

Conclusion

The Mars Rover is an **intelligent autonomous agent** that uses sensors to observe Mars, makes decisions based on its goals, and performs actions such as moving, collecting, and analysing samples. Since Mars is **partially observable, uncertain, and continuously changing**, a **model-based, goal-based, and utility-based agent architecture** is most suitable. Its performance can be measured by the **quality of exploration, samples collected, scientific data obtained, safety, and efficient use of resources.**

3. Explore the search space using search strategies and formulate the problem

components for the 8-Queens problem using the following information: Place 8

queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

8-Queens Problem – Problem Formulation and Search Space

Problem

Place **8 queens on an 8×8 chessboard** such that **no two queens attack each other**.

A queen can attack another queen if they are in the:

- Same **row**
- Same **column**
- Same **diagonal**

1. Initial State

- The chessboard is initially **empty**.
- No queens have been placed.
- Therefore, the initial state is:

0 queens placed on the board.

2. State

A state represents a **partial arrangement of queens** on the chessboard.

For example:

- [1] → Queen placed in column 1, row 1.
- [1, 3] → Queens placed in column 1, row 1 and column 2, row 3.
- [1, 3, 5] → Three queens placed without attacking each other.

A complete state contains **8 queens**.

3. Actions / Operators

The main action is:

Place one queen in an empty position such that it does not attack any previously placed queen.

For example:

1. Place Queen 1.
2. Place Queen 2 in a safe position.
3. Place Queen 3 in another safe position.
4. Continue until all 8 queens are placed.

4. Search Space

The search space consists of **all possible arrangements of queens** on the chessboard.

If we place exactly one queen in each column, each queen has **8 possible rows**.

Therefore, the maximum search space is:

$$\sqrt[8]{8^8 = 16,777,216}$$

However, the search algorithm **prunes invalid configurations** where queens attack each other.

5. Goal State

The goal is to place **8 queens** such that:

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

Example of a valid solution:

[1, 5, 8, 6, 3, 7, 2, 4]

This means:

Column	1	2	3	4	5	6	7	8
Row	1	5	8	6	3	7	2	4

6. Path Cost

- Each queen placement can be considered as having a cost of **1**.
- The total cost of a complete solution is **8 placements**.

7. Search Strategy

A common strategy is **Backtracking**, which is based on Depth-First Search.

Steps:

1. Start with an empty board.
2. Place a queen in the first column.
3. Move to the next column.
4. Check whether the position is safe.
5. If safe, place the queen.
6. Continue to the next column.
7. If no safe position exists, **backtrack** to the previous column.
8. Try another position.
9. Continue until all 8 queens are placed.

Conclusion

The **8-Queens problem** is a classic **search problem** where the objective is to find a configuration of 8 queens with no attacking pairs. **Backtracking/DFS** efficiently explores the search space by rejecting invalid configurations as soon as a conflict is detected.

5. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

OLA Cab Booking – Problem-Solving Agent

Type of Agent

The suitable agent is a **Utility-Based Agent**.

The customer has multiple choices such as:

- Mini
- Micro
- Sedan
- Shared
- Prime

The agent can compare different options based on **cost, comfort, travel time, distance, and availability** and select the most suitable cab.

Pseudocode

BEGIN

1. Get customer's current location.
2. Get customer's destination.
3. Display available cab types.
4. Get customer preference:
 - Cost
 - Comfort
 - Travel time
5. For each available cab:
 - Calculate fare
 - Calculate estimated travel time
 - Check availability
 - Calculate utility score
6. Compare the utility scores of all available cabs.
7. Select the cab with the highest utility score.
8. Display:
 - Cab type
 - Fare

Estimated arrival time

Estimated travel time

9. Ask customer for confirmation.

10. IF customer confirms

Book the selected cab.

Display driver and cab details.

ELSE

Allow customer to select another cab.

END

Example

Input:

- Source: College
- Destination: Airport
- Preference: Low cost + comfortable

Available:

- Micro → ₹300
- Mini → ₹350
- Sedan → ₹500
- Prime → ₹650

Decision: The utility-based agent compares the options and may select **Mini** because it provides a good balance between **cost and comfort**.

Conclusion

The **Utility-Based Agent** is most suitable because it evaluates multiple possible cab choices and selects the option that provides the **best overall utility according to the customer's preferences**.

5. logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform

Cost Search (UCS) algorithm.

Given Weighted Graph

From the diagram, the edge costs are:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

UCS Rule

Uniform Cost Search always expands the node with the lowest total path cost from the starting node S.

Step-by-Step UCS

Step 1: Start from S

Possible paths:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$

Choose **A** because cost **1** is smaller.

Step 2: Expand A

From A:

- $S \rightarrow A \rightarrow B = 1 + 3 = 4$
- $S \rightarrow A \rightarrow C = 1 + 1 = 2$

Choose **C** because cost **2** is the lowest.

Step 3: Expand C

From C:

- $S \rightarrow A \rightarrow C \rightarrow D = 2 + 1 = \mathbf{3}$
- $S \rightarrow A \rightarrow C \rightarrow G = 2 + 2 = \mathbf{4}$

Choose **D** because cost **3** is the lowest.

Step 4: Expand D

From D:

- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$
- $\text{Cost} = 3 + 3 = \mathbf{6}$

The available paths now include:

- $S \rightarrow A \rightarrow B = \mathbf{4}$
- $S \rightarrow A \rightarrow C \rightarrow G = \mathbf{4}$
- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = \mathbf{6}$
- $S \rightarrow G = \mathbf{12}$

The lowest cost is **4**.

Step 5: Reach Goal G

The path is:

$S \rightarrow A \rightarrow C \rightarrow G$

Total cost:

$$1 + 1 + 2 = 4$$

Step	Node Expanded	Path Cost
1	S	0
2	A	1
3	C	2
4	D	3
5	G	4

Final Answer

Least-cost path:

S->A->C->G

Minimum total cost = 4

Therefore, using **Uniform Cost Search**, the company should transport the package through **S → A → C → G**, with a minimum delivery cost of **4 units**.

Conclusion

Uniform Cost Search successfully finds the **least-cost path** by always expanding the node with the lowest total path cost. For the given delivery network, the optimal path is

S → A → C → G, with a **minimum total cost of 4 units**. Thus, UCS is suitable for finding the most economical route in a weighted delivery network.